

Watch-DevKit-HW

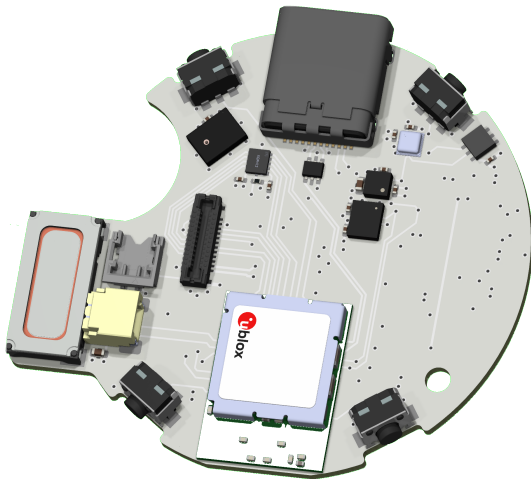
2026-02-20

Variant: CHECKED

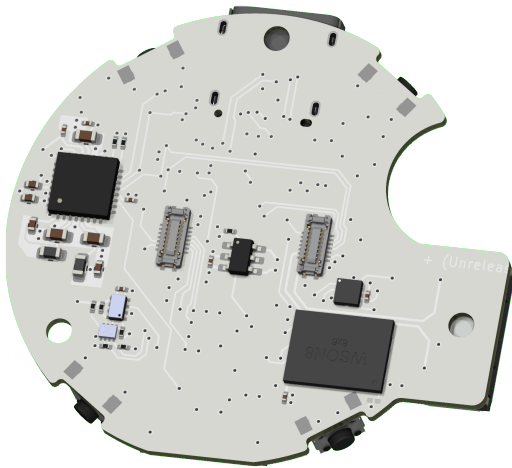
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 20-Feb-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for
informational design
notes.

DESIGN NOTE:
Example text for
debug notes.

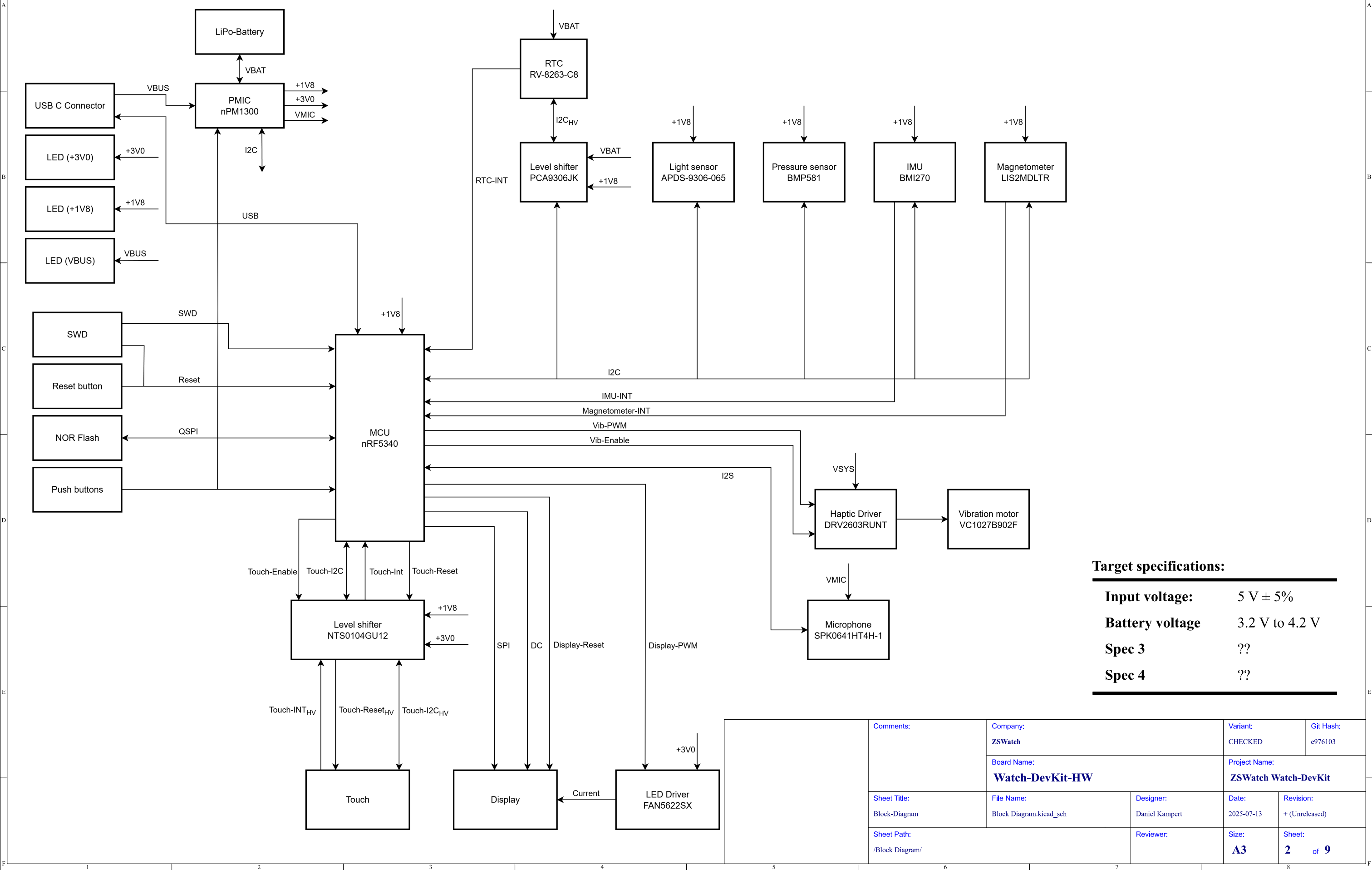
DESIGN NOTE:
Example text for
cautionary design
notes.

DESIGN NOTE:
Example text for
critical design
notes.

LAYOUT NOTE:
Example text for
critical layout
guidelines.

	Comments:	Company: ZSWatch	Variant: CHECKED	Git Hash: e976103
	Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Title: ZSWatch-Watch	File Name: ZSWatch-Watch.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13
	Sheet Path: /		Reviewer:	Revision: + (Unreleased)
			Size: A3	Sheet: 1 of 9

[2] Block-Diagram



	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

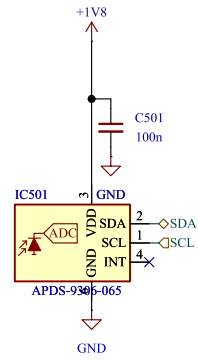
Target specifications:	
Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

	Comments:	Company: ZSWatch		Variant: CHECKED	Git Hash: e976103
		Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Title:	File Name: Power - Topology.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
	Sheet Path: /Power - Topology/		Reviewer:	Size: A3	Sheet: 2 of 9

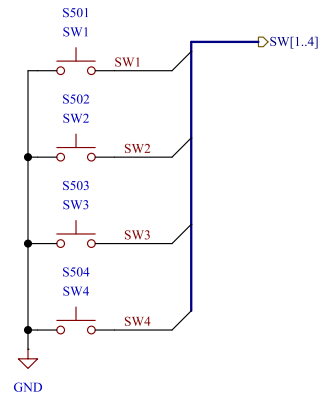


Comments:	Company: ZSWatch		Variant: CHECKED	Git Hash: e976103
	Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
Sheet Title: Power-Management	File Name: Power.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Power Management/		Reviewer:	Size: A4	Sheet: 4 of 9

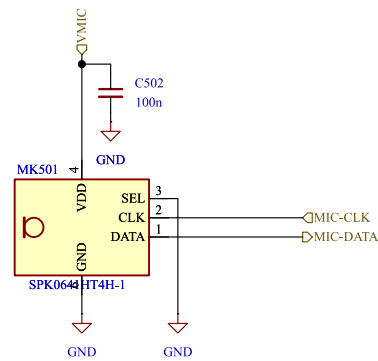
Light sensor



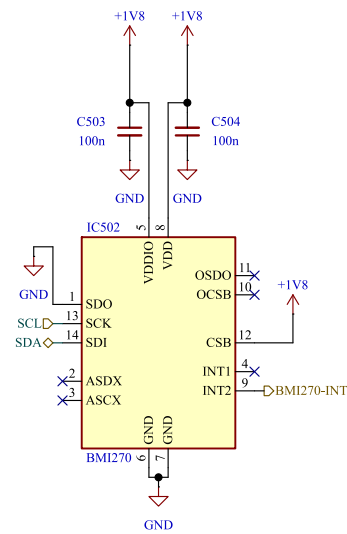
Buttons



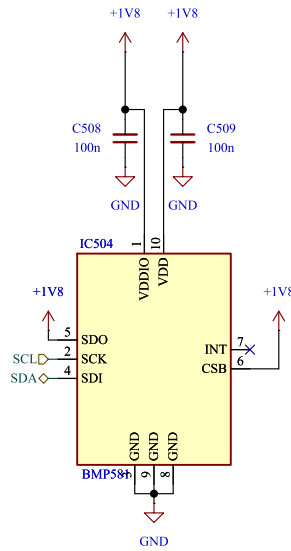
Microphone



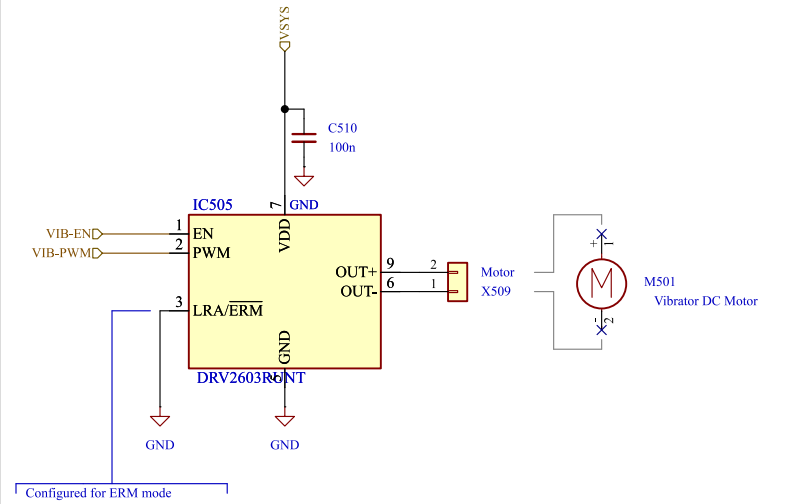
IMU with step counter and gesture detection



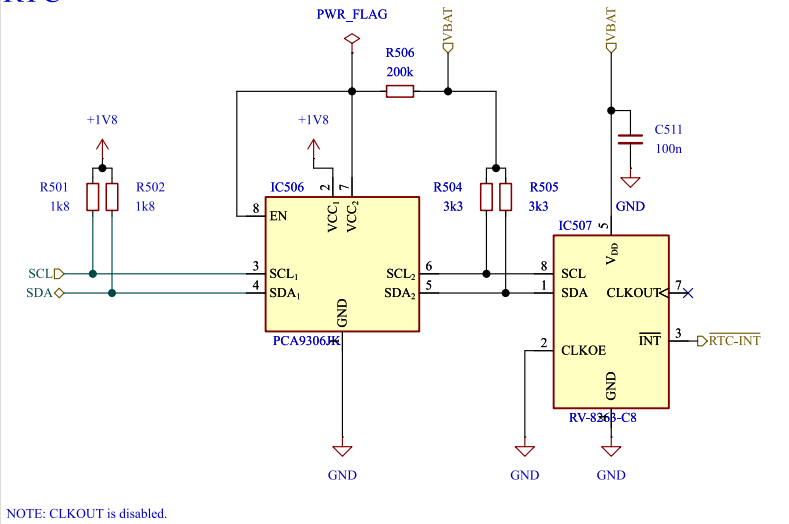
High accuracy pressure sensor



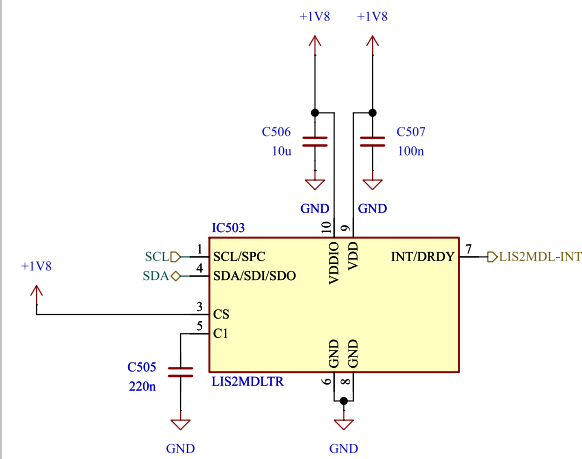
Vibration motor



RTC

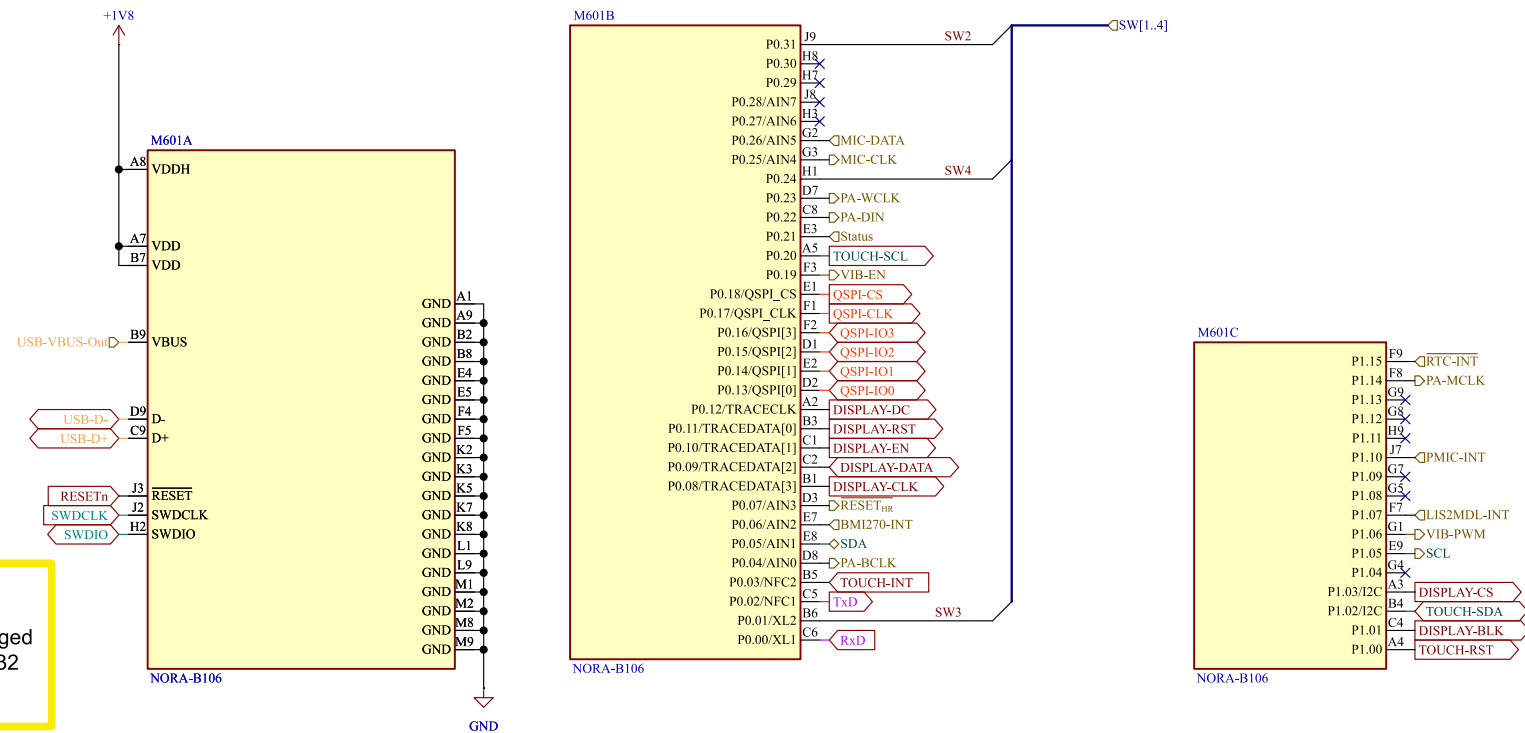


Magnetometer



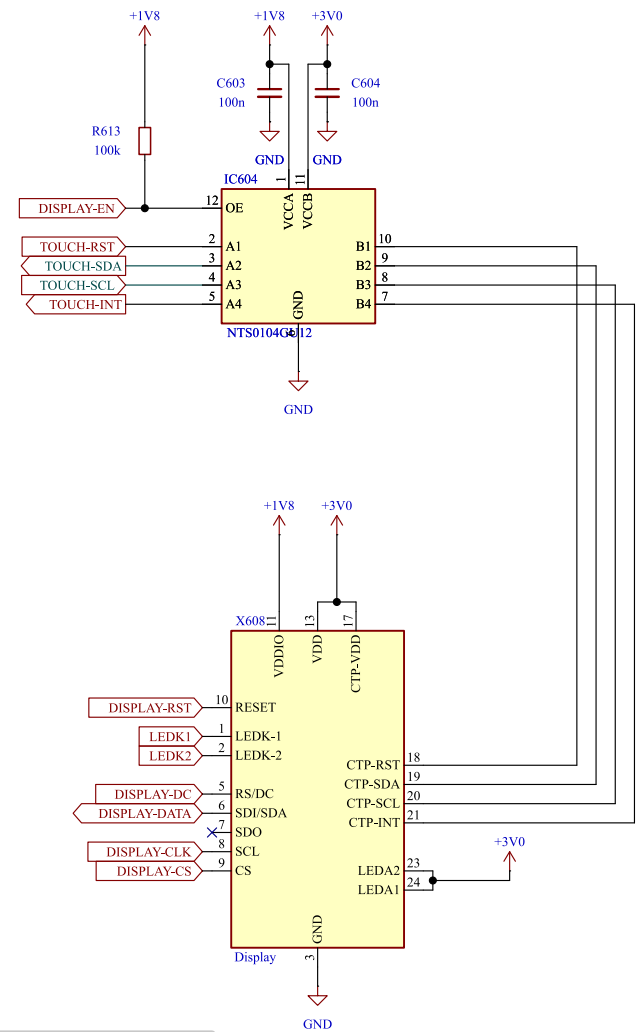
	Comments:	Company: ZSWatch		Variant: CHECKED	Git Hash: e976103	
		Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit		
	Sheet Title: Peripherals	File Name: Peripherals.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)	
	Sheet Path: /Project Architecture/Peripherals/		Reviewer:	Size: A3	Sheet: 5 of 9	

Microcontroller
(NORA-B106/nRF5340)



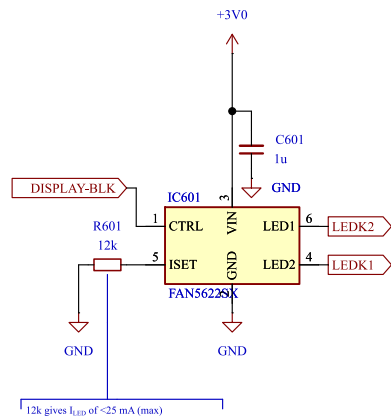
DESIGN NOTE:
The SPI lines for the display can't be changed and are required for 32 MHz operation.

Display



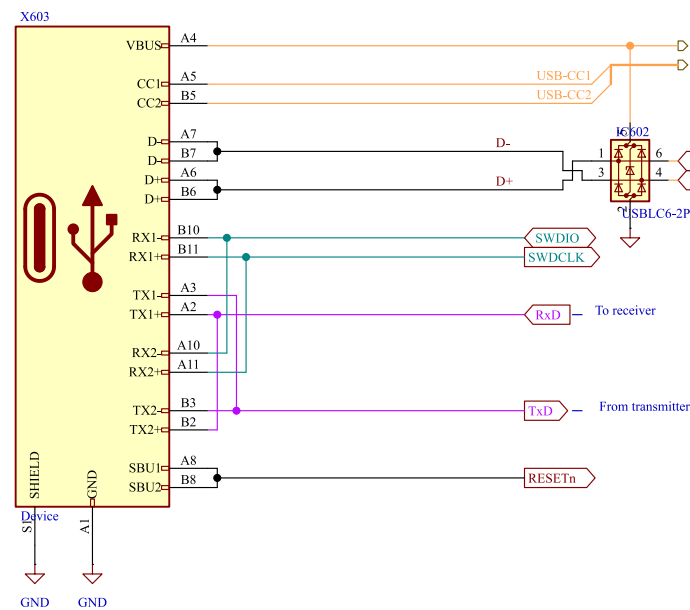
DESIGN NOTE:
NTS0104GU12 has
internal pullups.

Display backlight

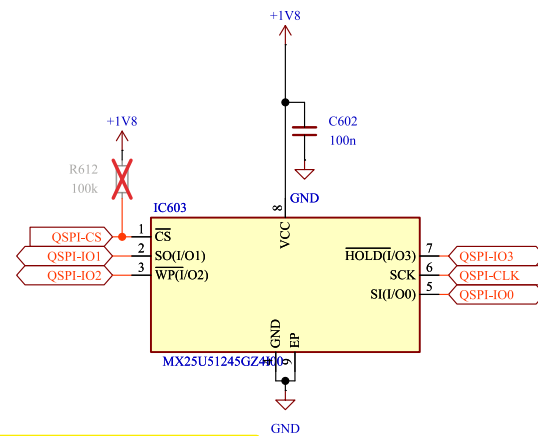


12k gives I_{LED} of <25 mA (max)

USB

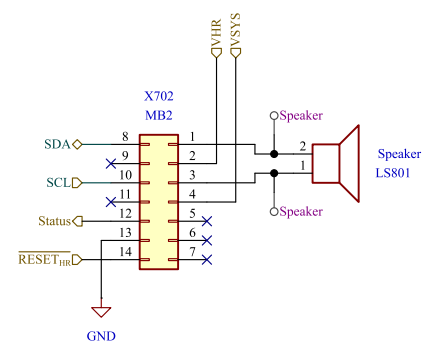
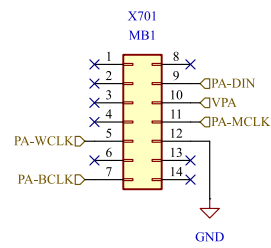


External Flash



DESIGN NOTE:
Make sure that the Flash
Memory supports
4k sector erase commands
(limited by QSPI driver)!

Comments:	Company: ZSWatch		Variant: CHECKED	Git Hash: e976103
	Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
Sheet Title: MCU	File Name: MCU.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
Sheet Path: /Project Architecture/MCU/		Reviewer:	Size: A3	Sheet: 6 of 9



	Comments:	Company: ZSWatch		Variant: CHECKED	Git Hash: e976103
		Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Title: Extension	File Name: Extension.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
	Sheet Path: /Project Architecture/Extension/		Reviewer:	Size: A4	Sheet: 7 of 9

[8] Revision-History

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	Comments:	Company: ZSWatch		Variant: CHECKED	Git Hash: e976103
		Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Title: Revision-History	File Name: Revision History.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: + (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 8 of 9